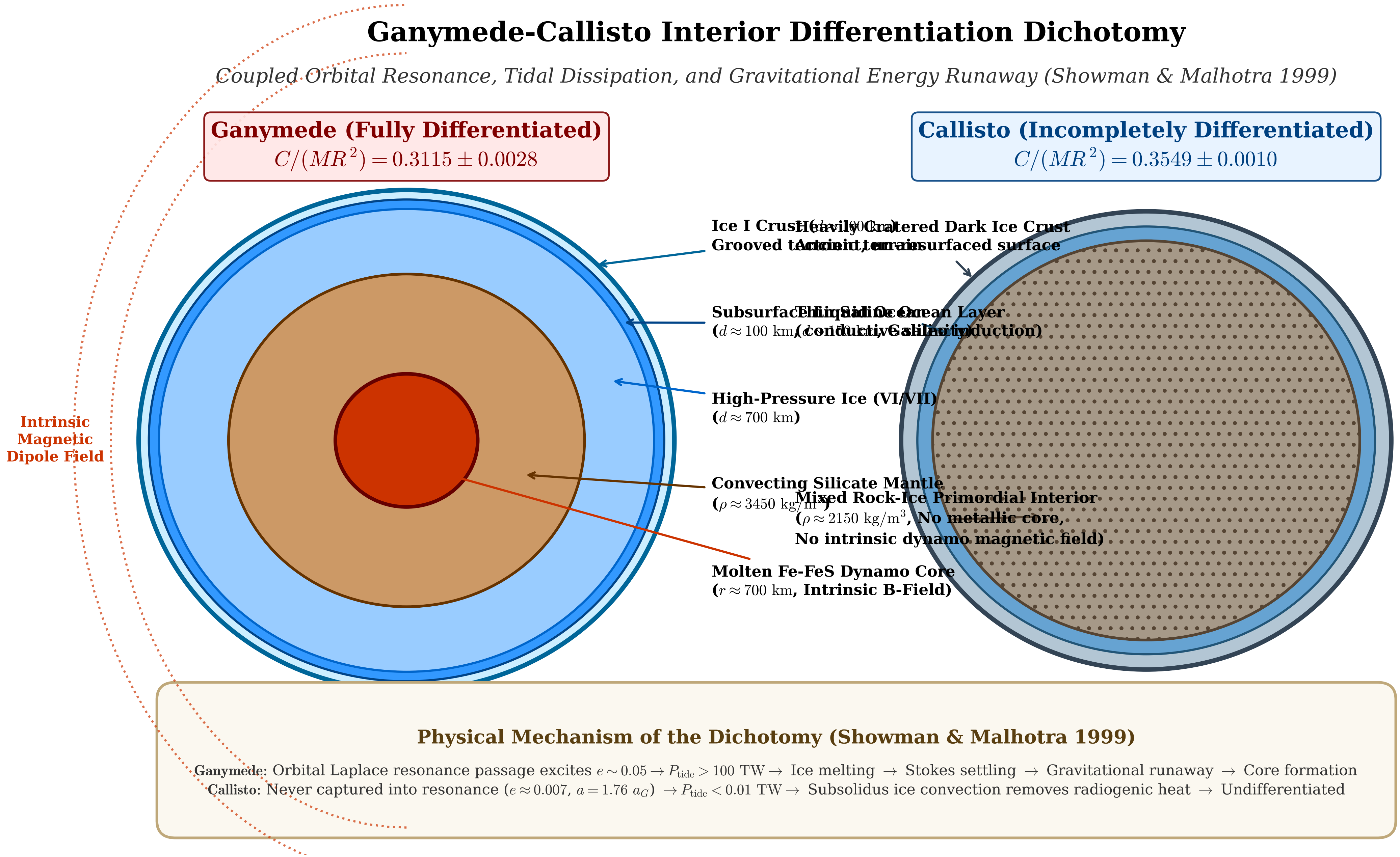
Ganymede-Callisto Interior Differentiation Dichotomy
Coupled Orbital Resonance, Tidal Dissipation, and Gravitational Energy Runaway (Showman & Malhotra 1999)
Ganymede (Fully Differentiated)
$C/(MR^2) = 0.3115 \pm 0.0028$
Callisto (Incompletely Differentiated)
$C/(MR^2) = 0.3549 \pm 0.0010$
Ice I Crust
Grooved tectonic terrain
Heavily Cratered Dark Ice Crust
Ancient, un-resurfaced surface
Subsurface Liquid Ocean
($d \approx 100$ km, conductive salinity)
Thin Saline Ocean Layer
Galileo induction)
High-Pressure Ice (VI/VII)
($d \approx 700$ km)
Convecting Silicate Mantle
($\rho \approx 3450$ kg/m^3)
Mixed Rock-Ice Primordial Interior
($\rho \approx 2150$ kg/m^3, No metallic core,
No intrinsic dynamo magnetic field)
Molten Fe-FeS Dynamo Core
($r \approx 700$ km, Intrinsic B-Field)
Intrinsic
Magnetic
Dipole Field
Physical Mechanism of the Dichotomy (Showman & Malhotra 1999)
Ganymede: Orbital Laplace resonance passage excites $e \sim 0.05 \rightarrow P_{\rm tide} > 100$ TW $\rightarrow$ Ice melting $\rightarrow$ Stokes settling $\rightarrow$ Gravitational runaway $\rightarrow$ Core formation
Callisto: Never captured into resonance ($e \approx 0.007$, $a = 1.76\ a_G$) $\rightarrow P_{\rm tide} < 0.01$ TW $\rightarrow$ Subsolidus ice convection removes radiogenic heat $\rightarrow$ Undifferentiated